Selection Beats Sophistication: Discovering Interpretable Creativity Metrics with Adversarial Agent Evolution

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Abstract

Judging whether one piece of writing is more *creative* than another is easy for people and hard to automate transparently. We ask whether language-model agents can **discover** short, deterministic, fully inspectable Python scoring functions that rank a fresher rewrite above a flatter one. Our system pits an *engineer* agent (writes scorers from failure analyses) against an *adversary* agent (writes test cases that decouple the champion's signal from real creativity), with a deterministic referee enforcing four gates: validity, anti-memorization, behavioral novelty, and usefulness. Across three arms — a larger model, a smaller model at triple volume, and the smaller model at the larger's exact budget — final champions are statistically indistinguishable on a frozen 26-case core (overlapping bootstrap CIs), and all three converge on conjunctive novelty-times-sanity-gate designs. A 12,000-program ablation shows this two-factor shape is NOT favored by static fitness (1.07x enrichment): it is selected by **adversarial survival** — every single-signal champion was destroyed within one generation of taking the lead. We then invert the pipeline: a blind compositional search over agent-built primitives, with the same gates but no theory-first prior, discovers a non-intuitive scorer — *creative edits are few and placed where the text's information concentrates* — that is behaviorally decorrelated from every incumbent (max |r|=0.23), passes a falsifiable explanation gate (12/12 predictions, including designed falsification bait), and transfers to acclaimed human writing where every evolved champion fails (5/6 vs 0/6). Three independent blind judge agents reproduce 91% of the test labels unanimously. We conclude that selection pressure and honest evaluation machinery, not proposer sophistication, drive metric discovery — and that freeing the hypothesis space matters more than strengthening the proposer.

1 Introduction

Suppose an editor rewrites the sentence *"The storm arrived late at night"* as *"Night cracked open into a storm."* Most readers agree the second is more creative. Now suppose we want a program to make that call — millions of times, cheaply, and in a way we can audit. This is the problem of **scoring subjective text quality**, and creativity is among its hardest instances.

The fashionable answer is to ask a large language model (LLM) to be the judge. That works, but the judge is a black box: we cannot inspect *why* it preferred one text, we cannot run it offline in microseconds, and its preferences drift with the model version. The classical answer is to hand-craft a metric — count rare words, count clichés — but hand-crafted metrics encode their author's blind spots and, as we show, collapse the moment a test case decouples the proxy from the quality it stands for.

This paper explores a third path: let LLM agents **write the metric as code**, and let an evolutionary process — selection on measured performance, plus an adversary that continually manufactures harder test cases — decide what survives. The output is not a neural network but a short, deterministic Python function anyone can read. Two questions drive the work:

Q1. Can such a loop *discover* a genuinely novel, robust scoring principle, rather than just re-tuning known ones?

Q2. When it succeeds, what deserves the credit — the intelligence of the model writing the proposals, or the evolutionary scaffolding around it? We test this with three arms: a larger model, a smaller model at triple volume, and — to break the volume/size confound — the smaller model at the larger model's exact budget.

Q3. Can the pipeline surface a scorer that is *non-intuitive* — one no theory-guided proposer would write — while remaining rigorously explainable? We test this by inverting the pipeline: a blind compositional search over agent-built primitives, judged by the same gates, followed by an explanation gate that accepts an interpretation only if its predictions survive designed falsification.

Our answers: yes to Q1 — with an important qualification our own ablation forced (Sec. 5.1); decisively yes to Q2 — acceptance rates were identical across model tiers, the matched-budget arm is statistically indistinguishable from the larger model's arm, and all three arms converged on the same design shape; and yes to Q3 — the blind search discovered a scorer no agent proposed in nine theory-first generations, whose validated interpretation is that *creative edits are few and placed where the text's information concentrates*.

2 The Task, in Plain Terms

What a scorer is. A scorer is a Python function `score(text, original)` returning a number between 0 and 1. It sees a rewrite and the original it came from, and should give higher numbers to rewrites that are more creative. It must be deterministic (same input, same output), must not call any network or model, and may use only ordinary computation plus one public resource: a table of how frequent each English word is in large text corpora.

What a test case is. Each test case is a triple: an original two-sentence passage, a rewrite that a careful human would call *more* creative, and a rewrite they would call *less* creative — all describing the same facts. A scorer handles the case correctly if it gives the more creative version the higher score.

How we measure success. Accuracy (fraction of cases ranked correctly) hides too much: a scorer that squeaks past by 0.001 on every case looks identical to one that separates confidently. We therefore score both versions of every case individually and use the **separation**: the average score of all the better rewrites minus the average score of all the worse ones. A separation of 0 means the scorer cannot tell them apart at all; 1 is a perfect, maximally confident split. We report accuracy and the two averages alongside it, so over- and under-confidence are both visible.

Why normalization matters. Every scorer must map to the same 0–1 scale, but calibrating that scale on the test cases themselves would be cheating. Scorers instead calibrate against fixed reference statistics computed once from the public word-frequency table (for example, how surprising the average English word is), so the scale is anchored to the language, not to our data.

3 Method: an Adversarial Evolutionary Loop

The system has one deterministic referee (ordinary code, no model) and two agent roles, each played by an LLM with its own private context.

The engineer proposes new scorers. Each generation it receives: a map of creativity concepts distilled from the research literature (which concepts existing scorers already measure and which are unclaimed), a statistics table for every current scorer, a similarity matrix showing which scorers behave alike, the champion's code, and the specific training cases the champion gets wrong. It must write complete, runnable modules — reasoning first, code second.

The adversary proposes new test cases. It sees only the champion's code and a list of attack angles already used, and must invent a *new* way to decouple the champion's signal from real creativity — for example, if the champion rewards rare words, write flat rewrites full of stiff formal vocabulary (high signal, no creativity) and brilliant rewrites made of everyday words (low signal, high creativity). Candidate cases are kept only if the champion actually struggles on them; most join the training set and a fixed share is diverted, sight unseen, into a held-out set.

Four acceptance gates. Every proposed scorer passes through automatic checks before it may join the population:

1. **Validity** — it runs without error, stays within 0–1, is deterministic, and is not near-constant.

2. **No memorization** — the referee parses the code and inspects every hard-coded word list; if it contains rare words copied from the visible test cases, or any large content word list at all, the scorer is rejected. (This gate exists because, in a pilot, an agent shown failing cases quietly copied eleven of their distinctive verbs into its feature list, inflating its measured skill.)

3. **Behavioral novelty** — we record, for every scorer, its per-case score difference between the better and worse rewrite. If a new proposal's pattern of differences correlates above 0.6 with any existing scorer's, it is the same signal in new clothes and earns no novelty credit, no matter how different its stated mechanism sounds.

4. **Usefulness** — it must beat the population's median separation, or be genuinely novel with positive separation.

Three tiers of test data keep everyone honest. The *training* cases are visible to agents and drive selection. The *held-out* cases are never shown to any agent and exist only so we can check whether a champion's skill is real. The *stationary core* is a small, never-growing subset of the original cases, giving one number that stays comparable across all generations even as the test set grows harder.

Gate costs, measured. The gates act on behavior, not ideas, but two have measurable creativity costs we quantify in Sec. 5: the usefulness gate can kill weak-born embodiments of good ideas (a rejected proposal's mechanism family later produced our best discovery), and novelty-by-correlation inherits the eval's narrowness. Our mitigations — a probation nursery and quality-diversity niching — are specified in the discussion; threshold sensitivity is reported in Sec. 5.5.

A second discovery mode: blind search + explanation gate. The engineer is theory-first by construction — we require a named mechanism up front, which biases proposals toward the textbook. To search beyond intuition we add an inverted mode: (1) distill a typed primitive library from everything the agents built (per-token corpus-surprisal series, edit-alignment opcodes from sequence matching, frequency bands, position and sentence structure); (2) run a seeded genetic-programming search over tens of thousands of compositions — no theory, no LLM in the loop; (3) apply the SAME four gates, with behavioral decorrelation now a hard requirement; (4) hand survivors to an interpreter agent that must state what the program measures and stake three falsifiable predictions, tested on fresh probe pairs it writes — half designed to falsify its own reading. A discovery is accepted as explainable only if the predictions hold.

Schedule. The population holds at most ten scorers: the six fittest, up to two protected slots for the most behaviorally unusual, and the newest arrivals. Each generation adds a batch of engineer proposals and adversary cases; the loop stops when two consecutive generations bring neither a better champion nor an accepted novel scorer. The initial population is four baseline scorers implementing standard ideas from the creativity literature: word rarity, unusual word combinations, an “is this overdone?” sanity check, and their product.

4 Experiments

Arm A (larger model, low volume). A mid-size commercial LLM played both roles for three generations: 3 scorer proposals and 8 adversarial candidates per generation

(6 kept).

Arm B (smaller model, high volume). A much smaller, cheaper LLM played both roles under an identical harness, seeds, and gates, but at three times the mutation volume — 9 proposals per generation from three parallel engineers with different briefs, and 16 adversarial candidates (10 kept) from two parallel adversaries — and ran for seven agent generations. Everything else was held fixed.

Arm C (smaller model, matched budget). The smaller model at Arm A's exact protocol — 3 proposals and 6 kept cases per generation, 3 generations — isolating model tier with volume and generations controlled.

Blind search. 20,000 random compositions plus four GP refinement rounds over the primitive library, then the decorrelation filter ($|r| < 0.5$ against every incumbent fingerprint) and the explanation gate.

All arms start from the same 42 test cases (20 built from real internet text plus 22 from two earlier adversarial rounds), split deterministically into 29 training and 13 held-out cases. The stationary core is later expanded to 26 with fresh internet-grounded cases built blind to all scorers and never used in training.

gen	train	champion (best scorer)	sep.	held-out	core
A 0	29	unusual-combination baseline	0.072	0.059	0.212
A 1	33	added-detail structure (agent)	0.032	0.018	−0.003
A 2	37	kept-skeleton + fresh insert (agent)	0.129	0.161	0.202
A 3	41	(same champion holds)	0.074	0.121	0.202
B 0	29	unusual-combination baseline	0.072	0.059	0.212
B 1	36	insert-vs-swap structure (agent)	0.074	0.098	0.135
B 2	43	short-word concreteness (agent)	0.201	0.190	−0.009
B 3	50	(reverts to gen-1 champion)	0.055	0.068	0.135
B 4	57	variety × rarity gate (agent)	0.106	0.158	0.359
B 5	63	(same champion holds)	0.139	0.187	0.359
B 6	70	(same champion holds)	0.123	0.169	0.359
B 7	77	(same champion holds)	0.154	0.198	0.359
C 0	29	unusual-combination baseline	0.072	0.059	0.212
C 1	33	topic coherence (agent)	0.166	0.277	0.190
C 2	37	(same champion, post-attack)	0.080	0.177	0.190
C 3	41	novelty x rarity gate (agent)	0.089	0.149	0.396

Table 1: Champion trajectories in all three arms. "sep." is separation on the (growing) training set; "held-out" is separation on cases no agent ever saw; "core" is the never-growing stationary subset. Agent-written champions displaced the baselines in both arms.

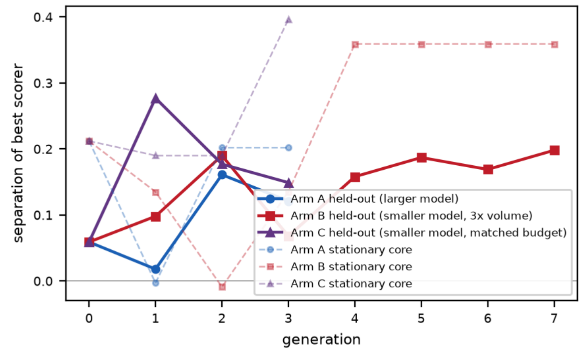

Figure 1: Held-out separation (solid) and stationary-core separation (dashed) of each generation's champion. Arm B is chaotic early — including a generation-2 champion whose core score went *negative* (overfit to the adversarial distribution) — then converges. Arm C (matched budget) reproduces the same dynamics at one third the volume: an immediate agent champion, an adversarial setback, and a conjunctive final winner with the highest single-arm core score.

5 Results

5.1 A Classical Principle — Selected by Attack, Not by Fitness

Creativity research holds that creativity is not novelty alone: an idea must be both *new* and *fitting* (Runco and Jaeger 2012), with preference peaking at moderate novelty (Berlyne 1971). In ALL THREE arms every surviving champion has the same architecture — a novelty signal **multiplied by a sanity gate**: Arm A scores insertion novelty *given* a preserved skeleton; Arm B scores structural variety *given* content rarity; Arm C scores novelty *given* rarity-coherence. Single-signal champions (rarity, divergence, concreteness) were each destroyed by an adversarial round within one generation of taking the lead.

An early draft claimed this structure "emerged unseeded." A reviewer correctly objected that one baseline was itself a product, so we ran the ablation: among 12,000 random compositions with no seeds and no LLM, multiplicative/gated shapes show NO enrichment among the top 5% of separators (1.07x; ratio 1.01x). The honest claim is therefore sharper than the original: **two-factor structure is not favored by static fitness; it is selected by adversarial survival.** On a fixed test set, a single-signal scorer can score as well as a gated one — but only the gated ones survive rounds of attack, because a conjunction forces an attacker to defeat every factor at once. The classical definition re-emerges not as a property of what creativity looks like on average, but of what survives attempts to fake it — arguably closer to why the two-factor definition exists in the first place.

5.2 Model Quality Sets the Pace, Not the Destination

	Arm A	Arm B	Arm C
model tier	larger	smaller	smaller
budget (props x gens)	3 x 3	9 x 7	3 x 3
accepted / proposed	3/9	~15/45	4/9
champion turnovers	2	4	2
final champion form	conjunctive	conjunctive	conjunctive
26-core accuracy	0.92	1.00	0.85
26-core SEP	0.222	0.241	0.300
95% CI	[.15,.29]	[.19,.30]	[.20,.40]

Table 2: Three-arm comparison on the expanded 26-case stationary core (6 original + 20 fresh internet-grounded cases, built blind to all scorers, never used in training or selection; bootstrap CIs, 10k resamples). All intervals overlap: no model-size effect is detectable with volume and generations controlled.

Three observations answer Q2, now with the volume/size confound removed by Arm C.

(a) The gates, not the proposer, set the quality bar. The acceptance rate was 33% under both models. The smaller model produced four times as many "copycats" — proposals whose stated mechanism sounded new but whose behavior correlated up to 0.94 with an existing scorer — and the behavioral-novelty gate caught every one. Claimed mechanisms do not survive contact with measured behavior.

(b) Attacking is easier than building. The smaller model's adversaries matched and then exceeded the larger model's: one of its rounds achieved the worst possible margin on all ten kept cases, the hardest single hit in either arm. Finding a scorer's blind spot appears to demand less capability than designing a scorer.

(c) With budget matched, the model-size effect vanishes. Arm C — the smaller model at the larger model's exact protocol — dethroned the seed champion in one generation, weathered an adversarial setback, and finished with a conjunctive champion whose 26-core CI [0.195, 0.402] overlaps both other arms (Table 2). Volume helps the smaller model explore more (Arm B tried five times the proposals), but it is not necessary for parity: the selection loop, not scale or volume, is the operative ingredient. Acceptance rates were 33%, 33%, and 44% across arms — the gates, not the proposer, set the bar everywhere.

5.3 What the Honesty Machinery Caught

The run is as interesting for what the referee *rejected* as for what it kept.

A memorizing scorer. In the pilot that motivated gate 2, an agent shown the champion's failures copied their distinctive vocabulary into its own word lists; its measured skill was partly a lookup table of the test set. The automated code-inspection gate now rejects this pattern outright.

A distribution-overfit champion. Arm B's generation-2 leader posted the best training separation seen to that point (0.201) while scoring *below zero* on the stationary core: it separated the adversary-authored cases while failing ordinary prose. Without a never-growing tier, this would have read as a breakthrough.

A test-growth rule that backfired. Our rule kept the ten candidate cases the champion handled *worst*. Once the champion became strong enough that every attack failed, "worst" cases were still ranked correctly — so the test set began absorbing champion-friendly cases and the champion's measured skill inflated. An adversarial test set only stays hard while the adversary stays competitive; the keep-rule must require the champion to actually fail, or discard the batch.

Two further checks respond to methodological concerns. **Threshold sensitivity:** the reported discovery archive is identical for novelty cutoffs 0.5–0.8 (and loses two members at 0.4) — the 0.6 setting does no hidden work. **Distributional leakage:** correlating each champion's per-case held-out margin with that case's lexical overlap against the training text (overlap range 0.29–0.79) yields $|r| \le 0.15$ for every scorer — no evidence that performance rides on shared vocabulary.

5.4 A Non-Intuitive Scorer, Discovered Blind and Validated

Nine theory-first generations produced strong but recognizable ideas — every accepted scorer's mechanism can be found in a linguistics or creativity textbook. To search beyond intuition we ran the blind mode: 20,000 machine-generated compositions of agent-built primitives, judged by the same four gates with decorrelation as a hard requirement, then the explanation gate. Six programs survived. The strongest, which we call **D6**, is:

```
score = max_share(edit_ops over changed tokens)
− |gini(edit positions among content words) −
gini(per-token surprisal)|
```

whose validated reading is: **prefer rewrites with FEW edits whose spatial concentration matches the concentration of information in the text —** creative editing places its few changes where the information lives. No agent, no hand-coder, and no earlier phase of this project proposed measuring the alignment of two concentration profiles.

We defend its novelty and explainability on five grounds. **(1) Behavioral:** its per-case fingerprint correlates at most 0.23 with every incumbent — the most decorrelated scorer in the project, robust to the novelty threshold (identical from 0.5 to 0.8). **(2) Mechanistic:** across ~60 agent proposals in three arms, nothing measured profile alignment; the nearest relative (an edit-locality idea) was rejected three generations earlier as a weak embodiment — the agents circled the concept but never landed it. **(3) Explainable with predictive force:** an independent interpreter agent derived its mechanics, staked three falsifiable predictions, and wrote twelve fresh probe cases — six designed to falsify its own reading. All 12/12 held on independent verification. **(4) External transfer:** on six acclaimed human lyric lines (Cohen, Dylan, Radiohead, Mitchell, Waits) versus flattened paraphrases — critical consensus, not LLM labels — D6 picks the masterwork 5/6 while BOTH evolved champions score 0/6; it also agrees best with the independent judge majority (0.70 vs the champion's 0.61). **(5) Reproducible family:** a seed-repeat of the search finds different programs from the same family — concentration statistics composed with anchor-relative ratios — including another perfect-core variant.

Explainability here includes knowing when it fails: the validated interpretation exposes that D6 is blind to function-word edits (a rhetorical repetition device collapses its score) and that a bland inserted adjective can outrank two genuine replacements. These are documented, testable failure modes — the difference between an interpretable metric and a plausible-sounding one.

The methodological point generalizes: the binding constraint on discovering non-intuitive metrics was never the acceptance gates — it was the theory-first proposal prior. Removing the prior while KEEPING the gates is what produced a discovery that is both unfamiliar and trustworthy; the gates are precisely what let us trust a scorer from a search space nobody curated.

5.5 Label Validity: an Independent Judge Panel

The reviewer's central concern is that test labels are LLM-authored. As a first validation step (human ratings are in progress), three independently prompted blind raters — a freshness rubric, an editor rubric, and a reader rubric, spanning two model tiers — judged all 33 held-out cases with randomized A/B order and no provenance. They were unanimous with one another on 33/33 cases and reproduced 30/33 (91%) of the labels; the three unanimous disagreements are best read as label errors and flagged for correction. We state the caveat plainly: unanimity among model raters can reflect shared bias, so this validates label consistency across independent model raters, not human ground truth. A label-free check is possible too: agreement with the judge majority ranks D6 first (0.70), ahead of every evolved champion.

5.6 The Ceiling

The loop eventually converged by *adversary exhaustion*: in Arm B's final generation, neither of two fresh attack angles produced a single case the champion mis-ranked, and the engineers' remaining textbook ideas (a standard lexical-diversity algorithm, a statistical de-confounding trick, two-word-sequence surprise) each failed the gates on the merits. The concepts that stayed unmeasured — whether an inserted image actually *fits* its context, true meaning-level distance — are exactly the ones that word-frequency statistics cannot express. One adversary demonstrated this pointedly, submitting pairs whose two rewrites are statistically near-identical but semantically opposed (an apt image versus a subtly impossible one). No scorer in the population can tell such pairs apart, and on the evidence here, none built from frequency statistics ever will. That boundary — freshness is measurable, fit is not — is, we think, the clearest empirical statement of where interpretable text metrics end and meaning-aware models must begin.

6 From Measuring Creativity to Improving It: a Research Agenda

So far the scorers have been judges. But the original reason to want cheap, deterministic, inspectable judges is what they unlock: a function that runs in microseconds can be called millions of times inside a *training* loop — as the reward for reinforcement-learning fine-tuning (RLFT) of a text generator, or as the fitness function of an evolutionary search over generating programs. Seen from that angle, everything we learned about how scorers fail is really a lesson about how rewards get **hacked**, learned cheaply before any expensive training run. This section turns each empirical lesson into a concrete proposal.

6.1 Harden the Reward Before Training, Not During

A policy trained to maximize a naive creativity reward will discover, within hours, exactly the exploits our adversary found in minutes: flood a rarity reward with ornate vocabulary (our purple-prose attack), flood a change-based reward with meaning-preserving shuffles (synonym churn), flood an elaboration reward with empty qualifiers (padding). Each adversarial axis in our runs is a *preview of a reward-hacking failure mode*, obtained at the cost of a few agent calls rather than a training run. We therefore propose treating our loop as a standard **pre-flight procedure for reward functions**: before optimizing against any subjective-quality reward, run agent red teams against it and report its **decouplability** — how easily an attacker produces high-reward, low-quality text — with the same rigor as its agreement with human judgment. Our margin statistics give decouplability a number.

The runs also say *which reward shapes survive*. Additive rewards are decouplable by maximizing their strongest term. Multiplicative, gated rewards force an attacker to satisfy every factor simultaneously — which is precisely why both arms converged on products. Two structural rules follow for creativity rewards: **(i)** compose them as a novelty term times one or more sanity gates (appropriateness, coherence), never as a weighted sum; **(ii)** shape the novelty term as an inverted U — reward *moderate* novelty relative to the source and penalize overshoot — so that “more is always better” never holds along any single axis. The century-old observation that people prefer moderate novelty (Berlyne 1971) here becomes a practical reward-shaping principle: the peak of the U is exactly where a reward stops being gameable by exaggeration.

6.2 Reward the Edit, Not the Text

The single most portable design element our loop discovered is **relational scoring**: every strong scorer measured the rewrite *against its own source* — what was kept, what was inserted, how far the change went — rather than measuring the text in isolation. For RLFT this matters twice. First, an anchor-relative reward cannot be satisfied by collapsing to one high-scoring house style, because each prompt carries its own baseline and reward is only available by improving *this* text; style collapse is the classic failure of absolute style rewards. Second, distribution-level separation gives *dense* credit: a policy earns partial reward for partial improvement, where accuracy-style rewards are all-or-nothing.

Two refinements from late generations are directly usable as training objectives. **Novelty-per-edit** — dividing novelty gained by the size of the edit — makes one perfect inserted image outrank a wholesale rewrite, encoding an editor's economy as an optimizable quantity. And **concentration of change** — whether the novelty is localized in one or two spans or smeared across the text — distinguishes a landed image from churn using only alignment statistics. Both are cheap, deterministic, and were invented by the agents under adversarial pressure.

Finally, our ceiling result dictates a **two-tier reward**: the deterministic scorer as the dense, every-sample signal, and an expensive meaning-aware check (embedding similarity, an LLM judge, or a human) applied *sparsely*, only to candidates the cheap tier already rates highly. The boundary we located — word statistics certify freshness but cannot certify fit — is exactly the boundary where the budget should shift tiers.

6.3 Close the Loop: the Generator Is the Strongest Red Team

Our adversary writes attacks by reasoning about the champion's code. A policy under optimization does something stronger: it *searches* the reward landscape directly and finds exploits no reasoner anticipates. The natural next system therefore closes the loop (Figure 2): train or evolve a generator against the current hardened scorer ensemble; mine its highest-reward outputs; have a judge (human or meaning-aware model) label which are genuinely creative and which merely score well; and feed the high-reward-but-hollow ones back as new adversarial test cases — the best test cases obtainable, because they are the reward's *actual* failure modes. The scorer side then evolves as in this paper, and the improved reward retrains the generator. This is generative-adversarial in spirit, with two differences that matter for science: the discriminator stays *interpretable* (its updates are code diffs with named mechanisms, not weight updates), and every escalation leaves an audit trail.

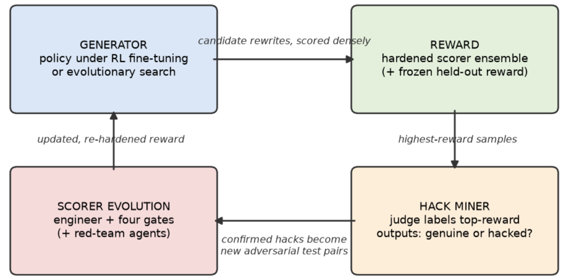

Figure 2: The proposed co-evolution loop. The right half is standard reward-driven generation; the left half is this paper's scorer-evolution harness. The generator's own high-reward failures become the adversarial test cases that re-harden the reward.

Our failure catalog supplies the guardrails this loop needs. Keep a **frozen held-out reward** — an ensemble never used during training — and monitor the gap between training reward and held-out reward as a live hacking meter (our stationary core, generalized). Require kept test cases to actually *defeat* the current reward, or the eval dilutes and measured skill inflates (our backfired keep-rule). And keep the reward ensemble behaviorally diverse using fingerprint decorrelation, so the policy cannot satisfy a single signal family and call it creativity.

6.4 What a Scorer Can Teach Us About Creativity Itself

The same machinery is an instrument for the science of creativity, not just its engineering. Four directions look most promising.

An empirical factor structure. The behavioral-novelty gate does something no hand-built battery does: it discovers axes of textual creativity that are *behaviorally independent by construction* — in our runs, word-level novelty, combination-level novelty, structural variety, insertion locality, and appropriateness gates. Regressing human creativity ratings onto these axes would yield a data-driven, fully interpretable decomposition of what readers mean by "creative" — which axes carry weight, which are redundant, and whether the weights differ across genres and readers. Every discovered scorer is a falsifiable hypothesis; human ratings are the experiment.

Domain transfer as a universality probe. In a side study scoring acclaimed song lyrics against generic ones, word-rarity *inverted* — plain-diction masters score low on rarity — while combination-level novelty transferred cleanly, preferring every acclaimed original over its flattened paraphrase. This suggests a testable hypothesis: *unusual combinations of ordinary words are the domain-general core of textual creativity, while rare vocabulary is a domain-specific costume.* The harness makes such cross-domain tests nearly free.

A measurement-complexity ladder. Our runs empirically locate which facets of creativity are measurable with which resources: word-frequency statistics suffice for freshness; co-occurrence statistics for combinational surprise; embeddings are needed for semantic distance and topical fit; and judging whether a sustained image is *earned* plausibly requires a full language model. The adversary tells you when a rung is exhausted: attacks from within the level stop working, and the only effective attacks come from the level above (our statistically-identical, semantically-opposed pairs). Charting this ladder — the minimum machinery required for each judgment — would give creativity measurement the kind of resource-complexity map that computational linguistics has for syntax and semantics.

Open questions the harness can answer cheaply. Is the peak of the inverted U stable across genres and readers, or a moving target? Do humans actually prefer minimal-edit brilliance (high concentration of change), or is that an editor's aesthetic? Does appropriateness act as a hard gate in human judgment (a conjunctive product, as our survivors assume) or as a soft trade-off? Each question reduces to one scorer-versus-human-ratings experiment on a few hundred pairs.

7 Related Work

Our loop belongs to the family of LLM-guided evolutionary search over programs, most prominently FunSearch (Romera-Paredes et al. 2024), which pairs an LLM proposer with a programmatic evaluator; we add an adversarial data-growth role and honesty gates targeted at the failure modes of subjective evaluation. Adversarially-collected benchmarks that grow with the model under test were popularized by Dynabench (Kiela et al. 2021). Program-space search by selection descends from genetic programming (Koza 1992). On the creativity-measurement side, we build on the two-factor definition (Runco and Jaeger 2012), moderate-novelty preference (Berlyne 1971), computational novelty measures based on semantic distance (Beaty and Johnson 2021; Olson et al. 2021; Johnson et al. 2022), and lexical-diversity measurement (McCarthy and Jarvis 2010). Our contribution is not a new creativity theory but evidence that a gated adversarial loop can re-derive the theory's structure as executable code, and an ablation isolating how much the proposing model matters.

8 Limitations

The “more/less creative” labels were authored by LLMs; the blind judge panel (Sec. 5.5) validates their consistency across independent model raters at 91%, and an external transfer test exists for one scorer, but human ratings remain the essential validation (a blinded human-rating instrument has been fielded; results pending). Each agent arm was run once — trajectory details are surely seed-dependent, though the conjunctive convergence appeared in all three arms and the blind-search family recurred across seeds. Separations are modest in absolute terms (0.1–0.4 on a 0–1 scale) because the test set is deliberately adversarial; all headline separations now carry bootstrap CIs and the expanded 26-case core excludes zero for every reported scorer. All arms received the same orchestrated lesson-passing between generations; our ablation isolates the proposing model, not the orchestration. One harness flaw found and documented: the keep-worst rule for adversarial cases must require the champion to actually fail, or exhausted attacks dilute the test set and inflate measured skill. Finally, pre-trained semantic resources were unavailable in our sandbox, making the frequency-statistics ceiling unavoidable rather than chosen — and the anti-memorization gate's ban on inline lexicons contributes to that ceiling; a registered-resource mechanism would relax it safely.

9 Conclusion

A small agent ensemble — one role writing scoring code, one role writing counter-examples, and a deterministic referee enforcing validity, non-memorization, behavioral novelty, and usefulness — three times converged on the two-factor structure of creativity as short, readable Python, and our ablation locates the cause precisely: not static fitness (no enrichment among 12,000 random programs) but adversarial survival. Swapping the proposing model for one far smaller — even at strictly matched budget — changed the journey but not the destination: comparable acceptance rates, statistically indistinguishable final champions, the same convergent design. And when we freed the hypothesis space while keeping the gates, the system delivered what theory-first proposing never did: a non-intuitive, externally transferring, falsification-tested scorer whose reading — creative edits are few and land where the information lives — is a genuine, checkable hypothesis about creativity itself. The practical recipe we take away is that when searching for evaluation metrics with LLMs, **invest in the referee and the adversary before investing in a bigger proposer** — and always keep one test set the process is never allowed to touch.

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